ACM Web Forge 2026

Event Report: AI-Assisted Development Workshops & Programming Jam

Prince Sultan University | College of Computer and Information Sciences | ACM Student Chapter

Academic Year 2026–2027, Fall Semester

Official event website: https://acmchapter-psu.github.io/acm-webforge-2026/

3	30	33	8	3,000
Workshops	Attendees per workshop	Jam registrants	Faculty judges	SAR in prizes



WebForge 2026 opening screen and event setup in the lab

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1. Executive Summary

ACM Web Forge 2026 trained up to 30 students per session in AI-assisted software development across three workshops, then put those skills to the test in a one-day Programming Jam on 31 October 2026. The event was organized and run entirely by the ACM Student Chapter at Prince Sultan University.

The workshops ran on 25, 26 and 28 October 2026 and moved participants from prompting an AI assistant for a first prototype, through AI pair programming for debugging and testing, to shipping a working full-stack web application. The Programming Jam drew 33 registrants across 31 entries from Software Engineering, Information Systems, Computer Science and other programs.

In the Jam, teams built service-booking web apps from a shared brief and were evaluated by a panel of eight CCIS faculty members. First place (1,500 SAR) went to **AlUla Site Tours**, a guided-trip booking app with conflict-free scheduling. Second place (1,000 SAR) went to **Bubble Quest**, an escape-room booking app. Third place (500 SAR) went to **SlotWise**, a service booking app.

> Web Forge extends the chapter's event series beyond cybersecurity. It sits alongside the semesterly PSU CTF and the ACM AI Hackathon as the chapter's development-focused track, and feeds directly into the CTF 3.0 cycle in November.

2. Event Identity and Strategic Rationale

Item	Web Forge 2026 Summary
Event name	ACM Web Forge 2026
Organizer	ACM Student Chapter, Prince Sultan University
College	College of Computer and Information Sciences (CCIS)
Format	3 hands-on workshops + 1 Programming Jam competition
Workshop dates	25, 26 and 28 October 2026
Competition date	31 October 2026
Venue	[TBD: room / lab]
Target audience	CCIS students at all levels; open to other colleges

AI coding assistants have become a standard part of how software is built, yet most coursework still treats them as off-limits or ignores them. Web Forge was designed to close that gap: to teach students to use AI tools as a productive, critical partner rather than a shortcut, and to give them a structured place to practise it.

The event also widens the chapter's reach. Earlier chapter events (PSU CTF 1.0 and 2.0, the ACM AI Hackathon) centred on cybersecurity and AI applications; Web Forge gives software engineering and information systems students a development-focused event of their own.

3. Objectives

1. Teach practical, responsible use of AI tools across the full development cycle: planning, coding, debugging, testing and deployment.
2. Build students' ability to review, verify and correct AI-generated code rather than accept it blindly.
3. Give participants a competitive setting to apply the workshop skills under time pressure.
4. Grow a cross-department community of student developers around the ACM chapter.
5. Produce portfolio-ready projects that students can show to employers and COOP placements.

4. Workshop Series

Three workshops ran on 25, 26 and 28 October 2026, each attended by up to 30 students. Each session built on the one before, so that by the third workshop participants had taken an idea to a deployed application.

Date	Workshop	Focus
25 Oct 2026	**Workshop 1:** Prompt to Prototype: Building Web Interfaces with AI	Writing effective prompts, scaffolding a project, generating and refining a responsive front end, reading and understanding AI output
26 Oct 2026	**Workshop 2:** AI Pair Programming: Debugging, Refactoring and Testing	Using AI to trace bugs, explain unfamiliar code, refactor safely and generate unit tests; spotting hallucinated or insecure code
28 Oct 2026	**Workshop 3:** From Localhost to Live: Shipping a Full-Stack App with AI Agents	Building a back end and API with an AI agent, connecting a database, version control and deploying to the web

Tools covered: Claude, Claude Code, ChatGPT, Google Gemini, GitHub Copilot, Cursor, Windsurf, v0 by Vercel, Bolt.new, Lovable and Replit Agent

Every session was hands-on: participants built alongside the presenter on their own laptops and left each workshop with working code.

5. Programming Jam Competition

The Programming Jam on 31 October 2026 was the capstone of Web Forge: teams applied the AI-assisted workflow from the workshops to build a working web application within a fixed time.

Item	Programming Jam Summary
Date	31 October 2026
Duration	4-hour build window, 10:00 AM – 2:00 PM (in person); judging 12:00 PM – 2:00 PM
Team size	Solo or teams of up to 4
Theme / problem	Application Brief 07: Service Booking App — an appointment-booking app for a small service provider and its customers, plus a hidden scheduling change request announced mid-competition
AI tools	Permitted and encouraged
Submissions received	10 teams evaluated
Evaluation	100-point rubric, scored by a panel of 8 CCIS faculty members

Competition tasks

Teams had to build a responsive web app where customers find open appointment slots, book a time and review their upcoming bookings, while a service provider views and manages the complete schedule. The core loop was Browse → Book → Manage on one shared schedule.

Required features

- **Authentication:** every booking is tied to the signed-in customer.
- **Availability:** once a slot is booked, it no longer appears as available to other customers.
- **Personal bookings:** each customer sees their own upcoming bookings.
- **Provider schedule:** a provider view shows bookings across all customers.
- **Persistent data:** bookings and account data survive a refresh and a new session.
- **Responsive UI:** core journeys work on mobile and desktop.

Optional stretch goals included cancel and reschedule, booking status labels, filters, and better accessibility, validation and loading/error states.

Required workflow and toolchain

Step	Stage	Tool
1	Understand and plan	Excalidraw system diagram
2	Design	Variant for UI/UX and key screens
3	Build	Visual Studio Code with Codex and/or Claude Code
4	Connect	Firebase authentication and database
5	Version	GitHub with meaningful commits
6	Deploy	Vercel
7	Configure	Custom domain through Cloudflare
8	Discover and measure	Google Search Console, Bing Webmaster Tools, PageSpeed Insights

> **Adaptability checkpoint.** Partway through the competition, a new client requirement changing one of the scheduling rules was announced. Teams had to assess its impact on screens, data, logic and permissions, adapt their app, and test it against both a conflict and a non-conflict case.

Final deliverables

- A deployed, working application URL
- A GitHub repository with meaningful commit history
- An Excalidraw system diagram the team can explain
- A live demo of the customer and provider journeys
- A short explanation of the implementation, key trade-offs and known limitations

Evaluation rubric

Judges scored each team out of 100 points using a structured scorecard and scoring guide, and asked for a live demonstration to verify key claims.

Category	What Judges Assessed	Live Check	Points
Functional Completeness	Customer sign-in, open slots, booking, personal bookings, provider schedule, booking integrity	Book a slot and verify it disappears for another user; refresh both views	20
Technical Implementation	Firebase authentication and persistence, permission controls, application logic, GitHub evidence	Show separate customer/provider access and persistent records; check commits	20
Problem Solving & Adaptability	Response to the announced scheduling change, debugging, keeping the existing flow working	Demonstrate the changed behavior with conflict and non-conflict cases	20
UI/UX & Design	Coherent navigation, clarity, responsive screens, accessible input and feedback	Open desktop and mobile views; check booking states and messages	15
System Understanding & Planning	Excalidraw diagram, data models, components, rationale and trade-offs	Explain relationships, authorization and slot-conflict prevention	10
Production Readiness & Performance	Live Vercel URL, Cloudflare domain, search setup, PageSpeed evaluation	Open the production URL; show domain, Search Console/Bing setup and PageSpeed result	10
Final Demonstration & Presentation	Clear, accurate, structured live demo and answers	Show both journeys, the adapted feature and key limitations	5
Total			**100**

Scoring principle: 0 = not implemented; partial points = incomplete or unreliable; full points = works correctly with verified evidence. Teams with equal totals share the same rank.

Judging panel

Dr. AbedAlRaoof Bsoul, Dr. Samah Almutlaq, Dr. Siwar Rekik, Dr. Sofianiza Abd Malik, Dr. Heyfa Ammar, Dr. Tariqullah Jan, Dr. Skander Turki and Dr. Sahar Amir.

6. Registration and Participation Statistics

33 students registered for the Programming Jam between 29 September and 25 October 2026, forming 31 entries. Each of the three workshops was attended by up to 30 students.

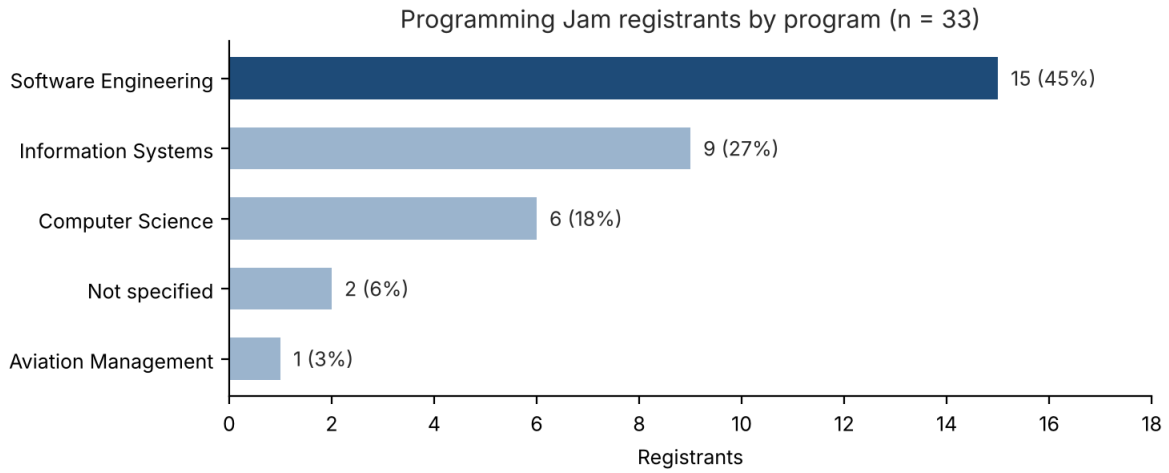


Figure 1. Programming Jam registrants by program (source: Web Forge registration form, 33 responses)

Registrations came from at least four programs, including one student from outside CCIS (Aviation Management). Sign-ups arrived in two waves: 10 in the first two days after announcement and 11 in the final four days before the workshops began. Two teams registered both members separately, and registrants named about 14 additional teammates on the form.

On competition day, 10 teams were evaluated by the judging panel.

7. Winners and Final Standings

The three winning teams each built a service-booking web app and shared a prize pool of 3,000 SAR.

Place	Project	Team Members (Name – ID)	What They Built	Award
1st Place	AlUla Site Tours	Abdulrehman Gelal – 224110470 Abdulmohsen Alghamdi – 223110119 Abdulaziz Ali Alghblan – 222110368	Booking app for private guided trips in AlUla. Time slots computed from operating hours, per-hour Firestore locks that prevent double-booking, atomic reschedule and cancel, role-based access, a live provider schedule and daily email reminders. React + TypeScript, Firebase, Vercel, Resend.	**1,500 SAR**
2nd Place	Bubble Quest	Lamar Alfakeh – 225411058	Booking app for The Maze, Escape Room and Mystery Lab: pick a day and time, bring up to 7 players and manage the booking online. React + Vite, Firebase, Vercel; live demo deployed.	**1,000 SAR**

Place	Project	Team Members (Name – ID)	What They Built	Award
3rd Place	SlotWise	Aljawharah Alruzuq – 223410216	Service booking web app built to the Jam brief, letting customers book and manage service appointments online.	**500 SAR**

8. Educational and Institutional Value

Web Forge created value for participants, organizers and the college. Students gained hands-on experience with tools now standard in industry, and the chapter demonstrated that it can run a multi-day development program end to end.

Impact Area	Evidence of Value
Industry-relevant skills	Students practised AI-assisted prompting, debugging, testing and deployment on real projects rather than isolated exercises.
Critical use of AI	Workshops stressed reviewing and verifying AI output, including spotting hallucinated or insecure code.
Applied learning	All three winning projects are working booking apps with authentication, a database and cloud hosting.
Cross-department reach	Registrants came from Software Engineering, Information Systems, Computer Science and a non-computing program.
Faculty engagement	Eight CCIS faculty members served on the judging panel.

9. Organizing Team and Appreciation

We extend our sincere appreciation to **Prince Sultan University** and the **College of Computer and Information Sciences** for their continued support of student-led technical initiatives.

We are especially grateful to **Dr. Mohammed Alshara, Dean of CCIS**, for his patronage and approval of the Fall 2026 event program, and to **Dr. Yasir Javed** for his continued guidance and support of the chapter's initiatives, from conception to execution.

Our thanks also go to the Programming Jam judging panel for their time and careful evaluation of the participating projects:

- **Dr. AbedAlRaoof Bsoul**
- **Dr. Samah Almutlaq**

- **Dr. Siwar Rekik**
- **Dr. Sofianiza Abd Malik**
- **Dr. Heyfa Ammar**
- **Dr. Tariqullah Jan**
- **Dr. Skander Turki**
- **Dr. Sahar Amir**

Special appreciation goes to the ACM Student Chapter team who planned and delivered the event, especially **Muhammad Yawar Hayat – 222211563 (President, ACM Student Chapter)** and **Shoug Alomran – 223410392 (Vice-President, ACM Student Chapter)**, who also delivered the workshops.

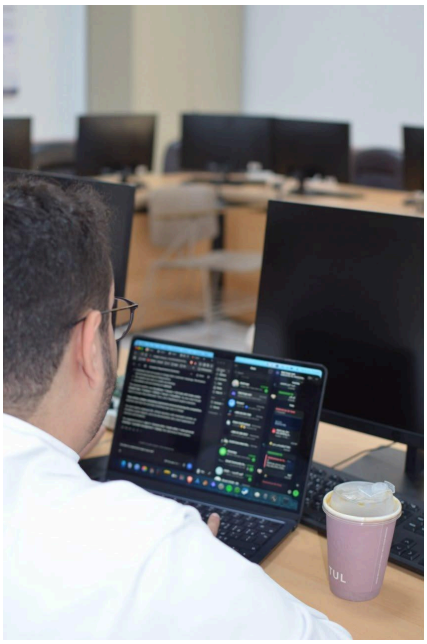
Finally, we congratulate all winners and participants. Their creativity and effort made Web Forge 2026 a proud achievement for the ACM Student Chapter, CCIS and Prince Sultan University.

10. Selected Photo Gallery

The following gallery records the workshops, the Programming Jam and the recognition of the winning teams.



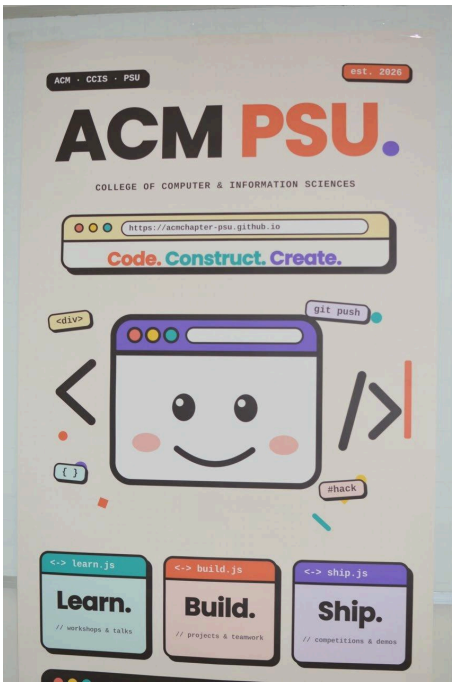
Participants with the ACM PSU "Smile, you're live" photo frame.



A participant building during the Programming Jam.



A team demonstrating its trip-booking app, including the reschedule flow, to the judges.



ACM PSU event banner: Learn. Build. Ship.

11. Recommendations for the Next Cycle

Recommendation	Purpose
Earlier registration with a mid-campaign reminder	Sign-ups came in two waves (10 in the first two days, 11 in the final four); a reminder would smooth demand.
One registration per team	Avoid duplicate entries and incomplete teammate lists.
Post-event feedback survey	Measure workshop quality and the Jam experience for the next edition.
Published materials and starter repositories	Let participants revisit the workshop content and build on it.
External sponsorship	Fund prizes and AI tool credits for participants.

Upcoming chapter initiatives

- **8, 9 and 11 November 2026:** PSU CTF 3.0 preparation workshops (web penetration testing and networking)
- **14 November 2026:** PSU CTF 3.0 competition
- [TBD: Spring 2027 plans]

12. Conclusion and Forward View

Web Forge 2026 delivered three hands-on workshops and a Programming Jam that drew 33 registrants from across CCIS and beyond. It gave students a structured, practical introduction to building software with AI, and established a development-focused event in the ACM chapter's annual program.

The chapter will build on this foundation in the CTF 3.0 cycle this November and in future editions of Web Forge.